Def. Given $n \in \mathbb{N}$, a polynomial

$$\Phi_n \overset{\text{def}}{=} \prod_{\xi \in \mu_n \text{ is generator of } \mu_n} (x - \xi)$$

is called $n^{th}$-cyclotomic polynomial.

Note: $\mu_n = \{\exp(2\pi i \cdot \frac{k}{n}) \mid k = 0, 1, \cdots, n-1\}$ is a group, under the multiplication.
Actually, $\mu_n \cong C_n$, the cyclic group of order $n$.
Moreover, for all $d \mid n$, $\mu_d \leq \mu_n$. Observe that

$$x^n - 1 = \prod_{\xi \in \mu_n} (x - \xi) = \prod_{d \mid n} \prod_{\substack{\xi \in \mu_d \\ \langle \xi \rangle = \mu_d}} (x - \xi) = \prod_{d \mid n} \Phi_d(x)$$

Thm. Let $\xi_n$ be the $n^{th}$ root of unity. Then,

$$(\mathbb{Z}/n\mathbb{Z})^\times \cong Gal(\mathbb{Q}(\xi_n)/\mathbb{Q}).$$

Proof. Let $(\mathbb{Z}/n\mathbb{Z})^\times \to Gal(\mathbb{Q}(\xi_n)/\mathbb{Q})$

$a \pmod n \mapsto \sigma_a$ where $\sigma_a(\xi_n) = \xi_n^a$.